Investigating the Heterogenous Impacts of Job Demands and Control: A Recentred Influence Functions Analysis of Employee Exhaustion*

Christina Boedker†
University of Newcastle

Maria Jahromi‡
Australian National University

Kieron J. Meagher§
Australian National University

March 25, 2025

Abstract

Karasek's (1979) original thesis on job demand-controls (JD-C) in the workplace suggested that job demands and job autonomy impact employees' emotional exhaustion. In the past 4 decades, a large number of studies into JD-C practices have followed. These have all been based on theoretical assumptions that employees are homogenous in their responses to workplace JD-Cs.

Contrary to these prior studies, in the current study we theorise that employees are heterogenous in their responses to workplace JD-Cs. Specifically, changes to job demands (hassles, work pressure, cognitive demands) and job autonomy impact employees' emotional exhaustion differently depending on their level of mental strain. This new approach of allowing for estimations of heterogenous impacts is made possible by recent developments in the recentered influence function.

We provide three empirical examples which shows the differential impacts of JD-Cs on employees' emotional exhaustion depending on their level of mental strain.

First, contrary to Karasek's original thesis that a combination of low job autonomy and heavy job demands is associated with high levels of exhaustion, we find that an increase in job autonomy provides no benefit to approximately one third of employees, being those with high mental strain. Increased job autonomy only reduces exhaustion for employees with low - medium mental strain (two thirds of the population) but not of employees with high strain, which paradoxically is the segment of employees that needs relief the most.

Second, whilst prior studies suggests that job demands increase employee exhaustion equally for all staff, we find that an increase in cognitive demands have

*Acknowledgements: We would like to thank ...

†Newcastle Business School, University of Newcastle, Newcastle NSW, Australia. *email:* Christina.Boedker@newcastle.edu.au

‡Research School of Economics, H.W. Arndt Building, Australian National University, Canberra, ACT 2601, Australia. *email:* maria.jahromi@anu.edu.au

§Research School of Economics, H.W. Arndt Building, Australian National University, Canberra, ACT 2601, Australia. *email:* kieron.meagher@anu.edu.au

opposing effects on exhaustion for high versus low strain employees. Emotional exhaustion reduces for approximately 40% of employees (those with low strain) but increases exhaustion for approximately 60% of employees (those with medium – high mental strain) when they experience higher cognitive demands.

Third, whilst work pressure and administrative hassles increase exhaustion for all employees, the effects for employees with high mental strain are half as strong when compared to those with low mental strain; suggesting that employees with high mental are so strained and depleted that they cannot mobilise a response to the additional job demands.

In conclusion, we contribute to extant literatures by showing that any workplace policy, which is uniformly applied by managers, will have differential, and at times contradictory, impacts on employees' level of emotional exhaustion. We suggest that any future research must consider how employees respond differently to changes in both job control and job demands depending on their prior levels of mental strain.

JEL classifications:
Keywords: Exhaustion, burnout, quantiles, recentered influence functions, job demands, job control.

1 Introduction

Workplace emotional exhaustion has been rising for decades, reaching its peak during the COVID-19 pandemic (Jenkins, 2023; Wilk et al., 2023; Sherman and Klinenberg, 2024). In Australia, about 46% of employees feel burnt out and 92% of workplace serious mental health issues are attributed to work-related stressors (ELMO Employee Sentiment Index report, 2022). Similar trends can be found in the US and the UK (ref.). Essential workers, including healthcare workers, are particularly vulnerable to experiencing higher burnout risk (Grace and VanHeuvelen, 2019; Haar and O'Kane, 2022; Jenkins, 2023). SafeWork Australia (2022) recently reported that work-related mental health conditions are on the rise, leading them to issue the Code of Practice for Managing Psychosocial Hazards under section 274 of the Work Health and Safety Act 2011. The Code mandates that Australian employers must honour their responsibility to identify, assess and eliminate workplace psychosocial hazards and risks, suggesting they are just as threatening to employees as physical hazards (Safe Work Australia, 2019, 2022). Legislations on psychosocial risk also exist in other countries, for example some European countries (Jain et al., 2022).

Psychosocial hazards can be defined as "all the aspects of management and work organization that may negatively affect the employee's mental and physical health" (Kowalczuk et al., 2018:1). In common with workplace bullying or psychological safety climate more broadly, workplace psychosocial hazards are largely influenced by an organisation's approach to leadership including the values, beliefs, and behaviours of supervisors (Dollard et al., 2017).

Psychosocial risks are recognised as a major challenge to occupational health and safety and public health more broadly (EU-OSHA (European Agency for Safety and Health at Work), 2007; Leka, S. and Cox, T., 2008). Exposure to psychosocial risks can lead to physical, mental and social health including burnout (World Health Organization, 2010). This means that high job demands, low control, and effort-reward imbalance are risk factors for mental and physical health problems (World Health Organization, 2010). It is therefore widely recommended that occupational health and safety also include work-related stress (for example, by the Commission for the Social Determinants of Health

(World Health Organization, 2008)).

The disparities and observed unevenness of the psychological hazards are particularly relevant to policy makers within governments (Sheehy et al. 2021), and in particular in the context of the Australian Work Health and Safety Act 2011 and the Australian Government's Mental Health Strategy (Safe Work Australia 2022), which place greater demands on employers' responsibility to identify and reduce psychosocial risks and hazards. The objective of the government's mental health strategy is to ensure that proactive measures are taken to identify, eliminate or minimise psychosocial risks in the workplace. The promotion of a positive workplace culture, for example, includes committing and supporting managers to develop their capability to create mentally healthy workplaces that promote a culture of positive mental health and well-being (Safe Work Australia, 2022).

Stressing that employers, including any person conducting a business or undertaking (PCBU), have a legal responsibility to manage hazards and risks in the workplace, the resources provided include access to a free account which encompasses five key risk management steps that guide users along the way. The steps are: 1) preparing the workplace; 2) conducting the survey (identifying risks); 3) understanding results (assessing hazards); taking action (controlling risks); and 3) reviewing and improving control measures (SafeWork Australia, 2021).

Emotional exhaustion has been shown to be a key factor that is detrimental to workers' satisfaction, engagement, performance, and intention to leave (e.g., Cropanzano et al., 2003; Halbesleben, & Buckley, 2004). However, as Moss (2019) suggests in focusing solely on emotional exhaustion, we may overlook the broader organisational issues that contribute to it, suggesting that a more systemic approach is necessary to address the underlying causes effectively. Variable pay systems (also known as performance-based systems) are associated with higher burnout compared to fixed salaries (Yeh et al., 2009).

The job demands-control (JD-C) model proposes that employees who face heavy job demands and low control over work are more likely to experience psychological distress (Karasek, 1979). [Add more literature here with similar research questions as ours but

using the standard approaches without accounting for the distribution]

Prior analyses of the impact of job demands and control on employee exhaustion have traditionally focused on average effects. We propose a framework allowing for heterogenous impacts, which vary depending on the level of quantiles of exhaustion. That is, how an employee responds to additional job demands or control varies depending on their level of exhaustion. This new approach is made possible by the recent development of recentered influence functions.

2 Methods

2.1 Participants and Sample

We use data from an Australian nationwide employee survey which includes a total of 2,638 observations. [add more details here including Ethics approval]

Data is pooled from three cross-sectional waves. The first wave was in the field between 8 and 29 December 2021, the second wave was conducted between 17 and 31 March 2022, and the third wave took place between 1 April and 11 July 2022.

We applied the following checks to determine the validity of the responses: (1) Straight line responses where respondents consistently score every question strongly agree / disagree; (2) The age of the respondent minus length of service was less than 15; (3) The average duration of a PhD degree (being 4 years) minus the respondent's age was less than 25; (4) The respondent's working hours was above 90; (5) The age of the respondent was greater than 75; (6) The number of the respondent's children was greater than 7; (7) The response time was less than 5 minutes. Any survey response that did not pass these checks was disqualified and deleted from the dataset. xx responses were rejected as invalid responses, leaving us with a dataset of xx valid responses, representing a response rate of x%. In addition, we restrict the sample to employees below the age of 67 who are located in Australia and we drop individuals with missing values. After applying these restrictions, the number of observations for each wave is 751, 604 and 620, respectively, or a total of 1,975 employees across all three waves.

2.2 Measures

2.2.1 Exhaustion

Emotional exhaustion is one of the key symptoms of burnout along with mental distance, emotional impairment and cognitive impairment (Maslach and Jackson, 1981; Haar and O'Kane, 2022). Among these burnout dimensions, emotional exhaustion is the closest to more orthodox job strain variables (De Jonge et al., 1999). We measure exhaustion using three items adapted from the emotional exhaustion component of Maslach Burnout Inventory (MBI) developed by Maslach and Jackson (1981). According to Maslach et al. (2001), the MBI is the scale with the strongest psychometric properties and it is widely used in the literature (De Jonge et al., 1999; Grace and VanHeuvelen, 2019; Hao and Zhang, 2024). We construct it via simple factor analysis from the following questions rated on an 11-point Likert scale from 1 to 11: (1) I feel burned out from my work. (2) I feel used up at the end of the workday. (3) I feel emotionally drained from my work. Each item was reverse coded so that higher scores were indicating higher levels of exhaustion. The internal consistency of the exhaustion scale was high, with Cronbach's alpha $\alpha = 0.90$.

2.2.2 Job demands

Job demands are captured via three constructs from Karasek Jr (1979) JD-C model that are widely used in the literature: work pressure (strain), cognitive demands and hassles (De Jonge et al., 1999; Elovainio et al., 2007; Oshio et al., 2015; Grace and VanHeuvelen, 2019). We use three-item scales for each construct with the items listed in Table 2.1. Higher scores indicate higher demands. Cronbach's alpha was $\alpha = 0.78$ for work pressure, $\alpha = 0.77$ for cognitive demands and $\alpha = 0.86$ for hassles.

Work pressure refers to the perceived intensity and urgency of job demands, often resulting in heightened stress when individuals feel they cannot manage these demands effectively (Karasek Jr, 1979). Prior literature uses work pressure as a proxy for job demands (Bakker, 2014; Grace and VanHeuvelen, 2019) and there are also variations of work pressure, such as time pressure (Demerouti et al., 2001), quantitative demands as one

of the dimensions in the Copenhagen Psychosocial Questionnaire (COPSOQ) (Kristensen et al., 2005) and work pace as one of the dimensions in the COPSOQ II (Pejtersen et al., 2010). Cognitive demands involve the mental effort required to process information, solve problems, and make decisions within the workplace, potentially leading to cognitive overload when these demands exceed an individual's capacity. Cognitive demands are one of the dimensions in the Copenhagen Psychosocial Questionnaire versions COPSOQ (Kristensen et al., 2005) and COPSOQ II (Pejtersen et al., 2010). Hassles relate to role conflict, ambiguity and overload (Zohar, 1997). Role clarity and role conflicts are also included as scales on the COPSOQ and COPSOQ II as part of the "Interpersonal relations and leadership" domain (Kristensen et al., 2005; Pejtersen et al., 2010).

[Note: terminology used in Social Science and Medicine: Work-life conflict (Grace and VanHeuvelen, 2019), irregular work hours (Grace and VanHeuvelen, 2019), heavy work pressure (Grace and VanHeuvelen, 2019), psychological (workload), physical and emotional demands (De Jonge et al., 1999). None of the SSM journal articles or the list of references on the linking document had "hassles"]

2.2.3 Job control

Job autonomy (decision authority or decisional influence) is commonly used as a proxy for job control (Blanch, 2016; Waddimba et al., 2019; Taouk et al., 2024). Job control is captured via three items on self-determination ($\alpha = 0.86$) which are derived from Spreitzer (1995) and are commonly used in the literature (Mahama and Cheng, 2013; Firth et al., 2014; D'Innocenzo et al., 2016; Dust et al., 2018). The three questions are listed in Table 2.1. Self-determination refers to an individual's sense of choice in initiating and regulating actions (Deci et al., 1989). It reflects autonomy in work behaviors, such as decisions on methods, pace, and effort (Spector, 1986). As a key component of psychological empowerment, self-determination, alongside meaning, competence, and impact, represents an active orientation to one's work role (Spreitzer, 1995).

[Note: The Social Science and Medicine journal uses decision authority or decisional influence rather than self-determination and I couldn't see any article that uses the Spre-

itzer scale.]

2.2.4 Factor analysis

All four variables are constructed via factor analysis from multiple questions that were asked in the survey. The factor analysis involves a novel approach whereby the factor analysis is applied to residuals from IV-type regressions to address potential measurement error. The regressions are estimated separately for every construct. Each individual variable within the construct is regressed on the other two or three variables that relate to the same construct, and then the residuals are extracted. The idea of using other measures of the same construct as IVs to reduce noise in the regression is based on Gillen et al. (2019), however our approach is novel in incorporating factor analysis into the scale construction.

Table 2.1: Constructs of Key Variables

Theoretical Context (Karasek, 1979)	Measurement Construct	Survey Items
Mental Strain	Exhaustion	(1) I feel burned out from my work. (2) I feel used up at the end of the workday. (3) I feel emotionally drained from my work.
Job Demand	Work pressure	(1) My job requires me to work very hard. (2) I have too much work to do. (3) I never seem to have enough time to get my work done.
Job Demand	Cognitive demands	(1) My work requires a lot of concentration. (2) My work is mentally very straining. (3) My work requires my constant attention.
Job Demand	Hassles	(1) I have to deal with a lot of administrative hassles in my job. (2) I have many hassles to go through to get my job done. (3) I have to go through a lot of red tape to get my job done.
Job Decision Latitude (Control)	Decision Authority (Autonomy)	(1) I have significant autonomy in determining how I do my job. (2) I can decide on my own how to go about doing my work. (3) I have considerable opportunity for independence and freedom in how I do my job.

2.2.5 Covariates

Covariates include eight state and territories, age, age squared, the number of employees in the organisation, tenure, tenure squared, whether born in Australia, Aboriginal and/or

Torres Strait Islander origin, education (university degree), presence of a long-term health condition, impairment or disability, the number of children, whether the employee is male, and work hours.

2.3 Descriptive statistics

Table 2.2 shows the summary statistics for all variables based on the final sample of 1,975 observations. Since the key variables are standardised, they all have a mean close to 0 and a standard deviation close to 1. Correlation matrices are shown for all key variables in Table 2.3 and for all continuous variables in Table 2.4. There does not appear to be multicollinearity as all correlations are below 55%. The highest correlation is 54% between exhaustion and work pressure. [Not sure if you wanted the correlation matrix for just the key variables or for all continuous variables so I included both versions. Correlation matrices are not ideal for categorical variables, so I haven't included those, but when I tried those in STATA there were no high correlations either].

Table 2.2: Summary Statistics

	Observations	Mean	SD	Skewness	Kurtosis	Min	Max
Exhaustion	1975	-0.01	1.01	-0.34	2.32	-2.06	1.60
Work Pressure	1975	0.00	1.01	-0.44	2.56	-2.29	1.59
Cognitive Demands	1975	-0.01	1.00	-0.66	3.42	-3.78	1.38
Hassles	1975	0.00	1.00	-0.34	2.62	-2.24	1.85
Autonomy	1975	0.02	1.00	-0.69	3.60	-3.77	1.52
Male	1975	0.38	0.49	0.48	1.23	0.00	1.00
Age	1975	34.83	10.99	0.80	3.03	18.00	66.00
Born in Australia	1975	0.79	0.41	-1.43	3.06	0.00	1.00
University Degree	1975	0.58	0.49	-0.33	1.11	0.00	1.00
Health Condition	1975	0.14	0.34	2.13	5.53	0.00	1.00
Number of Children	1975	2.01	1.17	1.52	8.91	0.00	12.00
Employees ('000)	1975	5.11	12.84	4.63	27.74	0.00	100.00
Tenure	1975	5.97	6.04	2.17	9.75	0.00	47.00
Work Hours	1975	40.37	14.77	0.22	4.69	0.00	99.00
ACT	1975	0.01	0.11	8.54	73.97	0.00	1.00
NT	1975	0.00	0.03	31.38	985.50	0.00	1.00
QLD	1975	0.21	0.41	1.44	3.08	0.00	1.00
SA	1975	0.05	0.22	4.05	17.42	0.00	1.00
TAS	1975	0.01	0.11	9.10	83.88	0.00	1.00
VIC	1975	0.25	0.43	1.16	2.35	0.00	1.00
WA	1975	0.08	0.27	3.07	10.43	0.00	1.00
Wave 1	1975	0.38	0.49	0.49	1.24	0.00	1.00
Wave 2	1975	0.31	0.46	0.84	1.71	0.00	1.00
Wave 3	1975	0.31	0.46	0.80	1.64	0.00	1.00

Table 2.3: Correlation Matrix of Key Variables

	Exhaustion	Work Pressure	Cognitive Demands	Hassles	Autonomy
Exhaustion	1.00				
Work Pressure	0.54***	1.00			
	(0.000)				
Cognitive Demands	0.26***	0.38***	1.00		
	(0.000)	(0.000)			
Hassles	0.45***	0.49***	0.29***	1.00	
	(0.000)	(0.000)	(0.000)		
Autonomy	-0.19***	-0.07***	0.18***	-0.07***	1.00
	(0.000)	(0.002)	(0.000)	(0.001)	

The sample includes 1975 individuals. p-values in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2.4: Correlation Matrix of Continuous Variables

	Exhaustion	Work Pressure	Cognitive Demands	Hassles	Autonomy	Age	Born in Australia	Children	Employees ('000)	Tenure	Work Hours
Exhaustion	1.00										
Work Pressure	0.54***	1.00									
	(0.000)										
Cognitive Demands	0.26***	0.38***	1.00								
	(0.000)	(0.000)									
Hassles	0.45***	0.49***	0.29***	1.00							
	(0.000)	(0.000)	(0.000)								
Autonomy	-0.19***	-0.07***	0.18***	-0.07***	1.00						
	(0.000)	(0.002)	(0.000)	(0.001)							
Age	-0.17***	-0.02	0.10***	0.00	0.10***	1.00					
	(0.000)	(0.500)	(0.000)	(0.949)	(0.000)						
Born in Australia	0.08***	0.06***	0.06**	0.01	-0.02	-0.16***	1.00				
	(0.000)	(0.009)	(0.011)	(0.583)	(0.330)	(0.000)					
Number of Children	-0.08***	0.01	0.08***	0.04*	0.09***	0.46***	-0.05**	1.00			
	(0.000)	(0.820)	(0.001)	(0.053)	(0.000)	(0.000)	(0.035)				
Employees ('000)	-0.00	0.01	0.00	-0.01	-0.00	0.04**	-0.01	0.01	1.00		
	(0.951)	(0.734)	(0.912)	(0.525)	(0.974)	(0.049)	(0.588)	(0.614)			
Tenure	-0.07***	0.01	0.07***	0.06***	0.11***	0.52***	-0.03	0.27***	0.16***	1.00	
	(0.003)	(0.559)	(0.001)	(0.006)	(0.000)	(0.000)	(0.198)	(0.000)	(0.000)		
Work Hours	0.11***	0.21***	0.09***	0.19***	0.08***	0.01	0.01	0.09***	-0.03	0.03	1.00
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.761)	(0.760)	(0.000)	(0.151)	(0.203)	

The sample includes 1975 individuals. p-values in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

2.4 Statistical analysis

Generate unconditional quantiles for the dependent variable.

All statistical analyses were performed with STATA software version MP 17. The significance level is 0.05.

3 Results

The main results are shown visually in Figure 1, while the estimated coefficients for all specifications are shown in Table 3.1. The figures show the relationship between each of the three demands and the control variables and exhaustion. The estimated coefficients and their respective confidence intervals are shown for quantiles 10 to 90 and illustrate how the relationship between those demands/control and exhaustion differs for individuals with lower levels of exhaustion compared to those with higher levels of exhaustion. The red lines indicate the average effect based on estimated coefficients from OLS regressions.

[Note on formatting Figure 1: Need to change self-determination to autonomy, white background, different y scales]

While experiencing work pressure and hassles increases exhaustion for individuals across all quantiles of exhaustion, these effects are greatest among individuals with lower levels of exhaustion. The effect is much lower for individuals with higher levels of exhaustion, with the effect reducing by about half for the 90th quantile relative to the 10th quantile. As shown in Table 3.2, these differences are statistically significant, as well as differences between the 10th quantile and the OLS estimate, and between the 90th quantile and the OLS estimate. This could be due to individuals disengaging as they are increasingly more exhausted and burnt out [check literature to back this up?].

A higher exposure to cognitive demands has the opposite effects for individuals with low versus high levels of exhaustion. Individuals with lower levels of exhaustion (quantiles 10, 20, 30) experience reductions in exhaustion when exposed to more cognitive demands. On the other hand, individuals who are already highly exhausted (quantiles 40 and higher)

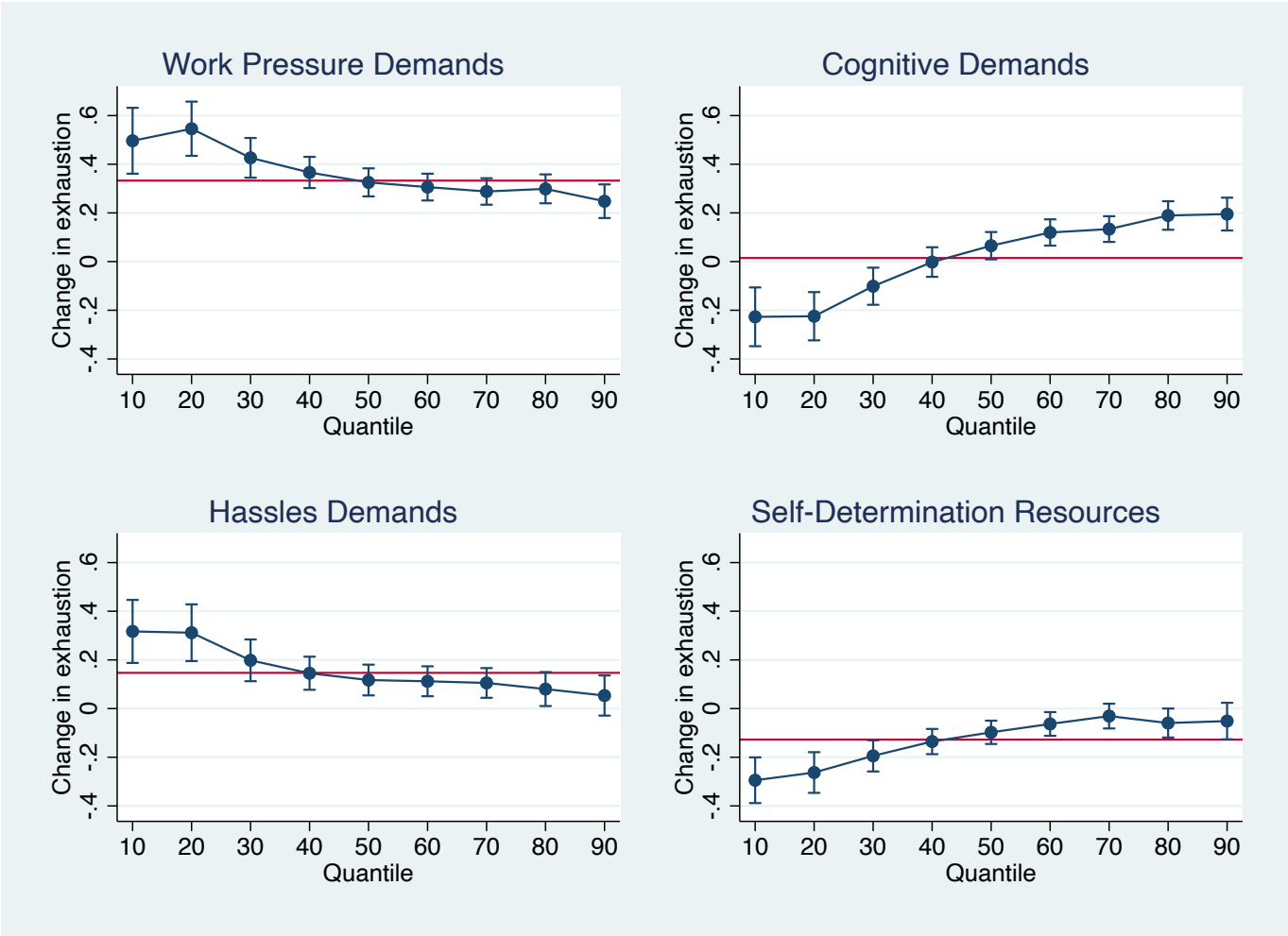

Figure 1: Coefficient Plots by Quantile with OLS Line

experience increases in exhaustion when exposed to more cognitive demands. This effect more than doubles at the 90th quantile relative to the 40th quantile. As shown in Table 3.2, these differences are statistically significant, as well as differences between the 10th quantile and the OLS estimate, and between the 90th quantile and the OLS estimate.

On the other hand, higher levels of autonomy reduce exhaustion for individuals across all quantiles of exhaustion, although this effect reduces for individuals who are already exhausted and becomes statistically insignificant for quantiles 70 and 90. Individuals who are already burnt out may have less capacity to benefit from autonomy [any literature?].

[Notes: change table 3.1 formatting to landscape long]

Table 3.1: Determinants of Exhaustion: OLS and Quantile Regressions

	(1) OLS	(2) Q10	(3) Q20	(4) Q30	(5) Q40	(6) Q50	(7) Q60	(8) Q70	(9) Q80	(10) Q90
Work Pressure	0.371**	0.539**	0.587**	0.498**	0.412**	0.365**	0.345**	0.325**	0.338**	0.278**
	(0.024)	(0.068)	(0.056)	(0.043)	(0.033)	(0.030)	(0.028)	(0.028)	(0.031)	(0.036)
Cognitive Demands	0.083**	-0.134*	-0.119*	-0.011	0.075*	0.141**	0.186**	0.192**	0.253**	0.248**
	(0.021)	(0.060)	(0.050)	(0.040)	(0.031)	(0.028)	(0.027)	(0.026)	(0.029)	(0.033)
Hassles	0.253**	0.447**	0.465**	0.363**	0.266**	0.241**	0.222**	0.195**	0.173**	0.129**
	(0.024)	(0.064)	(0.054)	(0.042)	(0.032)	(0.029)	(0.028)	(0.028)	(0.032)	(0.038)
Autonomy	-0.141**	-0.317**	-0.294**	-0.224**	-0.155**	-0.111**	-0.082**	-0.046	-0.072*	-0.059
	(0.019)	(0.048)	(0.043)	(0.035)	(0.027)	(0.025)	(0.025)	(0.026)	(0.031)	(0.038)
Male	-0.150**	0.048	-0.101	-0.178*	-0.150*	-0.185**	-0.201**	-0.232**	-0.296**	-0.253**
	(0.038)	(0.095)	(0.092)	(0.077)	(0.061)	(0.056)	(0.053)	(0.052)	(0.054)	(0.059)
Age	-0.021	0.029	-0.026	-0.037	-0.038*	-0.045**	-0.029	-0.030*	-0.014	-0.003
	(0.011)	(0.029)	(0.026)	(0.022)	(0.017)	(0.016)	(0.015)	(0.015)	(0.016)	(0.018)
Age Squared	0.000	-0.001	-0.000	0.000	0.000	0.000*	0.000	0.000	0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Born in Australia	0.054	0.119	0.044	0.075	0.042	0.019	0.057	0.044	0.050	0.111
	(0.045)	(0.111)	(0.106)	(0.089)	(0.071)	(0.064)	(0.062)	(0.062)	(0.067)	(0.065)
University Degree	-0.106**	-0.178*	-0.092	-0.062	-0.042	-0.053	-0.048	-0.134**	-0.140*	-0.188**
	(0.037)	(0.090)	(0.085)	(0.072)	(0.058)	(0.053)	(0.051)	(0.052)	(0.057)	(0.063)
Health Condition	0.070	-0.034	0.051	0.115	0.145	0.130	0.103	0.015	0.023	-0.012
	(0.053)	(0.119)	(0.116)	(0.098)	(0.081)	(0.075)	(0.073)	(0.075)	(0.082)	(0.089)
Number of Children	-0.023	0.016	-0.014	0.016	0.004	-0.025	-0.047	-0.037	-0.059*	-0.064*
	(0.017)	(0.040)	(0.038)	(0.034)	(0.027)	(0.024)	(0.025)	(0.023)	(0.025)	(0.029)
Employees ('000)	0.000	-0.007	-0.005	-0.004	-0.003	-0.001	0.002	0.004*	0.007**	0.006*
	(0.002)	(0.004)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
Tenure	0.014	-0.016	0.023	0.005	0.013	0.029**	0.031**	0.035**	0.025*	0.008
	(0.008)	(0.021)	(0.019)	(0.015)	(0.012)	(0.011)	(0.010)	(0.010)	(0.010)	(0.011)
Tenure Squared	-0.000	-0.000	-0.001	-0.000	-0.000	-0.001*	-0.001*	-0.001*	-0.001*	-0.000
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Work Hours	0.000	-0.001	-0.000	-0.001	-0.001	-0.001	-0.000	0.001	0.000	0.002
	(0.001)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
ACT	0.004	0.084	0.254	-0.402	0.001	0.158	0.106	0.039	-0.064	-0.262
	(0.122)	(0.361)	(0.313)	(0.293)	(0.225)	(0.194)	(0.213)	(0.204)	(0.216)	(0.187)
NT	1.235**	0.695*	1.030*	0.982*	0.984**	1.074**	1.328**	1.703**	2.387**	1.702
	(0.381)	(0.339)	(0.500)	(0.432)	(0.343)	(0.313)	(0.281)	(0.253)	(0.355)	(1.958)
QLD	0.047	0.061	0.120	0.095	0.126	-0.002	-0.017	-0.041	-0.071	-0.026
	(0.048)	(0.115)	(0.111)	(0.093)	(0.075)	(0.070)	(0.066)	(0.066)	(0.072)	(0.080)
SA	0.061	0.166	0.357*	-0.028	-0.001	0.102	0.112	0.088	0.080	-0.262**
	(0.079)	(0.198)	(0.182)	(0.163)	(0.127)	(0.117)	(0.115)	(0.120)	(0.130)	(0.101)
TAS	-0.084	0.459	-0.096	-0.084	0.008	-0.203	-0.299	-0.104	-0.365*	-0.253
	(0.161)	(0.428)	(0.434)	(0.350)	(0.276)	(0.236)	(0.196)	(0.201)	(0.183)	(0.194)
VIC	0.034	-0.077	0.110	0.020	0.005	0.009	0.088	0.034	0.019	0.028
	(0.046)	(0.113)	(0.104)	(0.088)	(0.072)	(0.065)	(0.063)	(0.063)	(0.069)	(0.078)
WA	0.049	0.203	0.152	0.040	0.069	0.119	0.047	-0.039	-0.102	-0.119
	(0.068)	(0.152)	(0.154)	(0.133)	(0.108)	(0.098)	(0.096)	(0.097)	(0.103)	(0.109)
Wave 2	0.033	-0.095	0.040	0.044	0.049	0.030	0.009	0.022	0.141*	0.038
	(0.046)	(0.109)	(0.103)	(0.087)	(0.070)	(0.065)	(0.063)	(0.062)	(0.069)	(0.079)
Wave 3	-0.057	-0.106	-0.027	-0.062	-0.038	-0.106	-0.125*	-0.058	-0.033	-0.100
	(0.043)	(0.106)	(0.102)	(0.086)	(0.069)	(0.063)	(0.059)	(0.060)	(0.064)	(0.069)
Observations	1975	1975	1975	1975	1975	1975	1975	1975	1975	1975
R^2	0.400	0.209	0.252	0.244	0.245	0.253	0.261	0.237	0.238	0.165

Table 3.2: Tests for Differences in Coefficients: 10th vs 90th Quantile - OLS vs 10th Quantile - OLS vs 90th Quantile

	10Q - 90Q	OLS - 10Q	OLS - 90Q
Work Pressure	F(1,1974) = 11.9 p = 0.001	F(1,1974) = 8.9 p = 0.003	F(1,1974) = 7.8 p = 0.005
Cognitive Demands	F(1,1974) = 29.6 p = 0.000	F(1,1974) = 18.4 p = 0.000	F(1,1974) = 28.0 p = 0.000
Hassles	F(1,1974) = 18.6 p = 0.000	F(1,1974) = 12.9 p = 0.000	F(1,1974) = 13.3 p = 0.000
Autonomy	F(1,1974) = 15.8 p = 0.000	F(1,1974) = 18.0 p = 0.000	F(1,1974) = 5.3 p = 0.021

4 Discussion

This section can include subsections on future research directions and limitations, theoretical and practical implications.

5 Conclusion

References

Bakker, A. B. (2014). The job demands–resources questionnaire. *Rotterdam: Erasmus University 10*, 9781118539415.

Bean, C. G., H. R. Winefield, C. Sargent, and A. D. Hutchinson (2015). Differential associations of job control components with both waist circumference and body mass index. *Social Science & Medicine 143*, 1–8.

Blanch, A. (2016). Social support as a mediator between job control and psychological strain. *Social Science & Medicine 157*, 148–155.

De Jonge, J., M. J. Mulder, and F. J. Nijhuis (1999). The incorporation of different demand concepts in the job demand-control model: effects on health care professionals. *Social Science & Medicine 48*(9), 1149–1160.

Deci, E. L., J. P. Connell, and R. M. Ryan (1989). Self-determination in a work organization. *Journal of applied psychology 74*(4), 580.

Demerouti, E., A. B. Bakker, F. Nachreiner, and W. B. Schaufeli (2001). The job demands-resources model of burnout. *Journal of Applied psychology 86*(3), 499.

Dust, S. B., C. J. Resick, J. A. Margolis, M. B. Mawritz, and R. L. Greenbaum (2018). Ethical leadership and employee success: Examining the roles of psychological empowerment and emotional exhaustion. *The leadership quarterly 29*(5), 570–583.

D'Innocenzo, L., M. M. Luciano, J. E. Mathieu, M. T. Maynard, and G. Chen (2016). Empowered to perform: A multilevel investigation of the influence of empowerment on performance in hospital units. *Academy of Management Journal 59*(4), 1290–1307.

Elovainio, M., M. Kivimäki, E. Ek, J. Vahtera, T. Honkonen, A. Taanila, J. Veijola, and M.-R. Järvelin (2007). The effect of pre-employment factors on job control, job strain and psychological distress: a 31-year longitudinal study. *Social science & medicine 65*(2), 187–199.

EU-OSHA (European Agency for Safety and Health at Work) (2007). Expert forecast on

emerging psychosocial risks related to occupational safety and health. Office for Official Publications of the European Communities.

Firth, B. M., G. Chen, B. L. Kirkman, and K. Kim (2014). Newcomers abroad: Expatriate adaptation during early phases of international assignments. *Academy of Management Journal 57*(1), 280–300.

Gillen, B., E. Snowberg, and L. Yariv (2019). Experimenting with measurement error: Techniques with applications to the caltech cohort study. *Journal of Political Economy 127*(4), 1826–1863.

Grace, M. K. and J. S. VanHeuvelen (2019). Occupational variation in burnout among medical staff: Evidence for the stress of higher status. *Social science & medicine 232*, 199–208.

Haar, J. and C. O'Kane (2022). A post-lockdown study of burnout risk amongst new zealand essential workers. *Social Science & Medicine 306*, 115157.

Hao, S. and X. Zhang (2024). Job burnout and anxiety among medical staff: A latent profile and moderated mediation analysis. *Social Science & Medicine 356*, 117141.

Jain, A., L. D. Torres, K. Teoh, and S. Leka (2022). The impact of national legislation on psychosocial risks on organisational action plans, psychosocial working conditions, and employee work-related stress in europe. *Social Science & Medicine 302*, 114987.

Jenkins, T. M. (2023). Physicians as shock absorbers: The system of structural factors driving burnout and dissatisfaction in medicine. *Social Science & Medicine 337*, 116311.

Karasek Jr, R. A. (1979). Job demands, job decision latitude, and mental strain: Implications for job redesign. *Administrative science quarterly*, 285–308.

Kristensen, T. S., H. Hannerz, A. Høgh, and V. Borg (2005). The copenhagen psychosocial questionnaire-a tool for the assessment and improvement of the psychosocial work environment. *Scandinavian journal of work, environment & health*, 438–449.

Leka, S. and Cox, T. (2008). The european framework for psychosocial risk management. Institute of Work, Health and Organisations.

Mahama, H. and M. M. Cheng (2013). The effect of managers' enabling perceptions on costing system use, psychological empowerment, and task performance. *Behavioral Research in Accounting 25*(1), 89–114.

Maslach, C. and S. E. Jackson (1981). The measurement of experienced burnout. *Journal of organizational behavior 2*(2), 99–113.

Maslach, C., W. B. Schaufeli, and M. P. Leiter (2001). Job burnout. *Annual review of psychology 52*(2001), 397–422.

Oshio, T., A. Tsutsumi, and A. Inoue (2015). Do time-invariant confounders explain away the association between job stress and workers' mental health?: Evidence from japanese occupational panel data. *Social Science & Medicine 126*, 138–144.

Pejtersen, J. H., T. S. Kristensen, V. Borg, and J. B. Bjorner (2010). The second version of the copenhagen psychosocial questionnaire. *Scandinavian journal of public health 38*(3_suppl), 8–24.

Safe Work Australia (2019). Work-related psychological health and safety: A systematic approach to meeting your duties. `https://www.safeworkaustralia.gov.au/system/files/documents/1911/work-related_psychological_health_and_safety_a_systematic_approach_to_meeting_your_duties.pdf`.

Safe Work Australia (2022). Model code of practice: Managing psychosocial hazards at work. `https://www.safeworkaustralia.gov.au/doc/model-code-practice-managing-psychosocial-hazards-work`.

Sherman, M. and E. Klinenberg (2024). Beyond burnout: Moral suffering among healthcare workers in the first covid-19 surge. *Social Science & Medicine 340*, 116471.

Spector, P. E. (1986). Perceived control by employees: A meta-analysis of studies concerning autonomy and participation at work. *Human relations 39*(11), 1005–1016.

Spreitzer, G. M. (1995). Psychological empowerment in the workplace: Dimensions, measurement, and validation. *Academy of management Journal 38*(5), 1442–1465.

Taouk, Y., Z. Aitken, A. D. LaMontagne, and T. King (2024). Persistent low job control and subsequent major depression: A prospective cohort study of australian working males. *Social Science & Medicine 359*, 117283.

Taouk, Y., M. J. Spittal, A. J. Milner, and A. D. LaMontagne (2020). All-cause mortality and the time-varying effects of psychosocial work stressors: A retrospective cohort study using the hilda survey. *Social Science & Medicine 266*, 113452.

Waddimba, A. C., D. C. Mohr, H. B. Beckman, T. L. Mahoney, and G. J. Young (2019). Job satisfaction and guideline adherence among physicians: moderating effects of perceived autonomy support and job control. *Social science & medicine 233*, 208–217.

Wilk, J. E., K. Clarke-Walper, K. Nugent, C. W. Hoge, M. Sampson, and C. H. Warner (2023). Associations of health care staff burnout with negative health and organizational outcomes in the us military health system. *Social Science & Medicine 330*, 116049.

World Health Organization (2008). Closing the gap in a generation: health equity through action on the social determinants of health: final report of the commission on social determinants of health. Commission on Social Determinants of Health.

World Health Organization (2010). Health impact of psychosocial hazards at work: an overview. `https://iris.who.int/handle/10665/44428`.

Yeh, W.-Y., Y. Cheng, and C.-J. Chen (2009). Social patterns of pay systems and their associations with psychosocial job characteristics and burnout among paid employees in taiwan. *Social science & medicine 68*(8), 1407–1415.

Zohar, D. (1997). Predicting burnout with a hassle-based measure of role demands. *Journal of organizational behavior: the international journal of industrial, occupational and Organizational Psychology and Behavior 18*(2), 101–115.

A Online Appendix

A.1 Definitions of psychosocial risks in the literature

Psychosocial job characteristics as assessed by Karasek's Demand-Control model: "Subjects were also asked to provide information regarding their working hours as well as their psychosocial work conditions. The average rate of working hours per week was measured by adding up all working hours during the week prior to the survey date. Shift work was measured by asking the subject if he/she was working shifts during the time period above. Psychosocial work demands (5 items) and job control (9 items) were assessed by the Chinese version of the Job Content Questionnaire (C-JCQ) based on Karasek's Job Strain model (Cheng, Luh, & Guo, 2003; Karasek & Theorell, 1990). This model postulates that a combination of high demands and low control causes high job strain and leads to negative health outcomes. The control dimension consists of two sub-components: decision authority and skill discretion. The model also postulates that demand-high control situations will enhance workers' active learning and lead to greater internal locus of control. Each item of the inventory was measured on a 4-point scale (strongly agree to strongly disagree). Detailed information regarding the psychometric properties of the C-JCQ can be found elsewhere (Cheng et al., 2003). In the present study, Cronbach's alpha coefficients for the job demands subscale were 0.31 for men and 0.39 for women, while those for the job control subscale were 0.87 for both men and women."

A.2 Definitions of burnout in the literature

Self-reported burnout status measured by the Chinese version of the Copenhagen Burnout Inventory: "Burnout status was assessed by the Chinese version of the Copenhagen Burnout Inventory (CBI) (Borritz & Kristensen, 2001; Kristensen, Borritz, Villadsen, & Christensen, 2005; Yeh, Cheng, Chen, Hu, & Kristensen, 2007). With CBI, burnout is conceptualized as a state of physical and psychological fatigue and exhaustion, and three dimensions of burnout are identified according to the life domain from which burnout may arise: (1) personal or generic burnout (6 items), which is designed to measure the degree of burnout experienced by the person (e.g., "How often do you feel tired?"), regardless of his or her employment status (including unemployed, early retired, students, housewives, etc.); (2) work-related burnout (7 items), which is designed to measure the degree of burnout perceived by the person at work as related to his or her work (e.g., "Do you feel burnt out because of your work?"); and (3) client-related burnout (6 items), which is designed to measure the degree of burnout perceived by the person as related to his or her work with clients (e.g., "Do you find it hard to work with clients?"), thus is suitable only for people whose work involve clients. The responses were recorded on a 5-point scale: from always (score 100) to never (0) or very seriously (100) to very slightly (0). In this survey, only the 'personal burnout' and 'work burnout' subscales of the CBI were used because of space limitation of the questionnaire. More information regarding the English version and the Chinese version of the CBI, including the details of all subscales, items and calculation formula, can be found elsewhere (Borritz & Kristensen, 2001; Yeh et al., 2007). In the 'work burnout' subscale, an item of the original C-CBId "Do you have enough energy for family and friends during leisure time?" was deleted in the analysis because it correlated poorly with the other items ($r < 0.1$). In the present study, Cronbach's alpha coefficients of the "personal burnout" subscale were above 0.93

and those of the "work burnout" were above 0.89 in both men and women." (Yeh et al., 2009)

"Burnout was measured using two validated instruments that assess current level of functioning. First was the Copenhagen Burnout Inventory (CBI) (Kristensen et al., 2005), a 13-item inventory that comprises the personal and work burnout scales. The core of the CBI measures physical and psychological fatigue and exhaustion. The CBI personal burnout scale is designed to measure overall burnout in an individual's life, while the work-related burnout scale measures burnout perceived to be related to work. This measure showed strong internal reliability (Cronbach's alpha = .85), and concurrent, predictive, and discriminant validity in samples of healthcare workers (Kristensen et al., 2005). Scores were calculated according to published criteria (Kristensen et al., 2005), where higher scores indicate greater burnout. In previous studies with U.S. healthcare workers, moderate-high levels of work-related burnout on the CBI have been seen (Thrush et al., 2021). Quartiles were also created based on the range of scores in the sample, with the top 25% considered to have high levels of burnout. The second measure of burnout was the Mini-Z (Linzer et al., 2001). The Mini-Z is a one-item screener that asks, "Using your own definition of 'burnout,' please select one of the answers below." Response options are on a 1–5 scale, ranging from "I enjoy my work. I have no symptoms of burnout." to "I feel completely burned out. I am at the point where I may need to seek help." Scores of ≥ 3 are considered indicative of burnout. The Mini-Z allowed comparisons to the 2019 Patient Safety Culture Survey (2019 Culture of Patient Safety Survey – Army, unpublished data, 2019) which used this measure." (Wilk et al., 2023)

Job burnout is an essential topic for researchers and a pressing issue for employers and employees. However, the most popular tool has become widely critiqued, and a new measure of burnout – the Burnout Assessment Tool (BAT) – is used here. The BAT is helpful because it provides a cut-off threshold score representing high burnout risk: "Job Burnout was measured using the 23-items of the BAT scale (Schaufeli et al., 2020a, 2020b), coded 1 = never to 5 = always. The four core symptoms of burnout are exhaustion (8-items, sample "At the end of my working day, I feel mentally exhausted and drained" $\alpha = 0.91$), mental distance (5-items, sample "At work, I do not think much about what I am doing and I function on autopilot" , $\alpha = 0.86$), emotional impairment (5-items, sample "At work, I feel unable to control my emotions" , $\alpha = 0.91$), and cognitive impairment (5-items, sample "I'm forgetful and distracted at work" , $\alpha = 0.93$). Confirmatory factor analysis (CFA) in AMOS (version 26) endorsed the higher-order model: χ^2(df) =1158.9(229), CFI = 0.94, RMSEA = 0.07, and SRMR = 0.06. Following the logic of Hadzibajramovic et al. (2020), we combined the dimensions to create the job burnout scale ($\alpha = 0.96$). Burnout Risk was calculated from job burnout (above), following Schaufeli et al. (2020a, 2020b) using Rasch analysis (for more details see Hadzibajramovic et al., 2020) to establish a cut-off level of Mean ≥ 3.30 (1 = above threshold, 0 = below, representing high burnout risk. For Hypothesis 3, we also split this into a risk for essential/non-essential workers." (Haar and O'Kane, 2022)

"To assess the degree of burnout, the Maslach Burnout Inventory-General Survey (MBI-GS) was administered (Schaufeli et al., 1996). The Chinese version of MBI-GS, which is a 15-item seven-point Likert scale ranging from 0 (never) to 6 (every day), was developed by Li and Shi (2003). The sub-dimensions of the scale are emotional exhaus-

tion, cynicism, and professional efficacy. Emotional exhaustion and cynicism are scored in the positive way, and professional efficacy is scored in the reverse way. The higher the MBI-GS score, the higher the burnout level. In the current study, Cronbach's alpha coefficient for the total scale is 0.856." (Hao and Zhang, 2024)

"Burnout captures feelings of emotional and mental exhaustion rooted in the recurrent experience of workplace stress. A modified burnout scale is adapted from the emotional exhaustion component of Maslach's Burnout Inventory (Maslach et al., 2001; Schaufeli et al., 1996). Participants were asked the extent to which they agreed with the following statements: "I feel emotionally drained from my work," "I feel used up at the end of the workday," "I feel fatigued when I get up in the morning and have to face another day on the job," "I feel burned out from my work," "I feel I'm working too hard on my job," and "I feel like I'm at the end of my rope" (=.82). Response categories for these questions range from (1) strongly disagree, to (5) strongly agree. A summary scale for burnout was created for all respondents (range=6–29)." (Grace and VanHeuvelen, 2019)

Emotional exhaustion: "This variable was assessed with the Spanish version of the Maslach Burnout Inventory (Maslach and Jackson, 1986, 1997). The scale has 9 items and measures feelings of being emotionally drained and exhausted by one's work (I feel emotionally drained from my work; I feel that I am working too hard on my job). The frequency of the respondents experience was assessed with a seven-point response format from never (0) to every day (6). Higher scores indicate higher levels of emotional exhaustion." (Blanch, 2016)

"Emotional exhaustion is measured by four items of the Dutch version of the Maslach Burnout Inventory: the MBI-NL (Schaufeli and van Dierendonck, 1993, 1994). Emotional exhaustion is the burnout-dimension that is closest to more orthodox job strain variables (cf. Shirom, 1989; Maslach, 1993). The questionnaire has a five-point response scale (ranging from 1 'never' to 5 'always'). An example item is: "I feel emotionally drained from my work". A study by Mulder et al. (1997) among 3,718 Dutch workers reported an alpha of 0.85 for this particular scale." (De Jonge et al., 1999)

A.3 Definition of job control

"The scale of job control (Karasek et al., 1981) comprised nine items (e.g., "My job allows me to make a lot of decisions on my own", "I have a lot of say about what happens on my job", "My job requires a high level of skill", "My job requires that I learn new things", "I get to do a variety of different things on my job", and "I have an opportunity to develop my own special abilities"). Responses were given along a five-point Likert-type scale ranging from 1 = strongly disagree to 5 = strongly agree. The job control index was the sum of response scores (Cronbach's alpha reliability $\alpha = 0.87$)." (Elovainio et al., 2007).

"Job control was evaluated with two subscales, skill utilization (6 items) and decision authority (3 items). Skill utilization taps whether a worker can apply his or her abilities to job performance (My job requires that I learn new things; My job requires me to be creative). Decision authority measures a worker capacity to make decisions at work (My

job allows me to make a lot of decisions on my own). Items were scored on a four-point Likert type scale, from 1 (strongly disagree) to 4 (strongly agree). High scores indicated higher levels of job control." (Blanch, 2016)

"Job control was assessed as the aggregate of three items pertaining to decision authority and two items pertaining to skills discretion rated on a 7-point Likert scale from 0 (strongly disagree) to 6 (strongly agree). The three decision authority items included:'I have the freedom to decide how I do my own work';'I have a lot to say about what happens on my job'; and'I have a lot of freedom to decide when I do my work' (Cronbach's alpha=0.83). The two skills discretion items included: My job often requires me to learn new skills'; and 'I use many of my skills and abilities in my current job' (Cronbach's alpha=0.68). All five items were reversed coded so that higher scores were indicative of lower job control. The scores for each subscale were calculated by summing the responses for the relevant item in each subscale, and total scores for decision authority and skill discretion summed to calculate the overall job control score. Summary scores ranging from 0 to 30. Summary scores were stratified into tertiles, as in previous studies (Taouk et al., 2021). Those in the highest tertile with scores ranging from 15 to 30, reflecting the most adverse conditions, were categorised as "exposed," while the remaining two tertiles were amalgamated to establish the reference category. A measure of persistent low job control was established by examining consecutive low job control exposure across two waves (Wave 1 and Wave 2). Instances where job control fluctuated between low and high control across two waves were classified as intermittent exposure, and continuous high job control exposure over both waves was categorised as persistent high job control." (Taouk et al., 2024)

"We defined individual job control as the perception by each physician that he/she had sufficient mastery over the processes and resources necessary, plus adequate decisional influence (e.g. overpatients' health-related behavior), to accomplish the financially-incentivized clinical tasks. (...) For job control, candidate items from the Job Content Questionnaire's decisional latitude subscale (Karasek et al., 1998) and from procedural justice dimensions (e.g. process control, decision control) of an organizational justice measure (Colquitt,2001) underwent a similar process to create the "perceived control" scale ($\alpha = 0.69$;p $< .001$)(Meterkoetal.,2006;Waddimbaetal.,2010). This scale comprises three items (e.g. "Actions necessary to obtain the financial incentive are within my control"). Participants rated their agreement with each item on both scales using a five-level Likert-type format ranging from 1=strongly disagree to 5 = strongly agree. Because our hypotheses were directed to the high and low ends of the distribution for the two moderator variables (i.e.,focusing on physicians who reported high (or low) autonomy support and/or high job control), we conducted analyses by dichotomizing the scale scores for autonomy support and job control. Specifically, we coded average ratings below 3 as "low autonomy support" or "low job control"and ratings equal to or above 3 as "high" values." (Waddimba et al., 2019)

A.4 Definition of job demands

"Job demands scale (Karasek et al., 1981) comprised four items. (e.g., "My job requires working very fast.") Responses were given along a five-point Likert-(type) scale ranging

from 1 = strongly disagree to 5 = strongly agree. The job demands index was the sum of response scores (Cronbach's alpha reliability $\alpha = 0.82$)." (Elovainio et al., 2007).

"Emotional labor. The extent to which respondents engage in surface and deep acting are measured by items adapted from Krumland Geddes(2000).Surface acting items ask respondents the extent to which they agree (1=strongly disagree, 4=strongly agree) with the following statements: "I show feelings to parents that are different from what I feel inside," "I have to cover up my true feelings when dealing with parents," and "I fake the emotions I show when dealing with parents." These items were combined to form a mean surface actingscale(=.73).Four items are used to measure deep acting. These items ask respondents to assess the degree to which they agree (1=strongly disagree, 4=strongly agree) with the following statements: "When working with parents, I try to create certain emotions in myself to present the image the hospital desire," "I talk myself out of feeling what I really feel when helping parents," "If I think the hospital would not approve of my real feelings about parents,I try to change those feelings,"and "When helping parents,if I pretend I'm happy, I can actually start to feel it." These items were similarly combined to form a mean deep acting scale (=0.72)." (Grace and VanHeuvelen, 2019)

"Work-life conflict. Work-life conflict is measured by an item that asks, "How often does your job interfere with your family life?" Responses range from (1) practically never, to (5) almost always." (Grace and VanHeuvelen, 2019)

"Irregular work hours. Respondents 'frequency of irregular work hours is assessed by a question that asks, "How often do you have to work irregular hours?" Responses span from (1) practically never, to (5) almost always." (Grace and VanHeuvelen, 2019)

"Heavy work pressure. Respondents' experience of work pressure is gauged by an item that asks, "To what extent are you constantly under heavy pressure at your job? "Responses range from(1) not at all, to (5) to a very great extent." (Grace and VanHeuvelen, 2019)

Job demands and control: "We utilized the items investigating job demands and control from the Japanese version of the JCQ. The JCQ is based on the JD-C model, and includes scales related to job demands (five items) and job control (nine items) rated on a four-point scale ($1\frac{1}{4}$ strongly disagree to $4\frac{1}{4}$ strongly agree; Karasek, 1985). The internal consistency, reliability, and validity of the Japanese version of the JCQ have been shown to be acceptable (Kawakami et al., 1995). In the present sample, Cronbach's alpha coefficients were 0.69 and 0.78 for the job demands and control scales, respectively. We summed the responses to these items into single indices of job demands (range: 12-48) and control (range: 24-96). We also calculated the job demands/control ratio to measure the extent of job strain (Landsbergis et al., 1994). Finally, we used the means as the cut-off points for the binary variables that classified each worker as having either high or low job demands, job control, and job strain." (Oshio et al., 2015)

"The Job Content Questionnaire (JCQ) (Karasek et al., 1998) was used during a computer-assisted telephone interview to collect data on work-related psychological demands, skill discretion, decision authority, coworker support and supervisor support. The JCQ is the most commonly used instrument to capture the JDCS model variables (Cour-

voisier and Perneger, 2010) and has established reliability and validity (de Araújo and Karasek, 2008; Karasek et al., 1998). Psychometric properties including internal reliability estimates are provided in Table 2. The version used in this study contained 35 items with a 4-point response scale (e.g. 1 = "strongly disagree", 4 = "strongly agree"). In order to build indicators for each subscale of the JDCS model, a sum of the weighted item scores was calculated according to the JCQ user guide (Karasek, 1985)." (Bean et al., 2015)

"We used three different types of demands (i.e. psychological, physical and emotional demands) and job autonomy as predictor variables. The psychological job demands and job autonomy measures differ somewhat from Karasek's original measures, because we have tried to operationalize these constructs more precisely and more completely (cf. Wall et al., 1996). Furthermore, to minimize bias, the subjective indicators of our job characteristics contain items with a minimum of cognitive processing. In other words, these items are precisely defined and are as neutral as possible (cf. Frese and Zapf, 1988; Wall et al., 1996). Finally, response categories are frequency-based rather than intensity-based grading (cf. Wall et al., 1996). (De Jonge et al., 1999)
Psychological job demands (i.e. workload) are measured by an eight-item questionnaire (five-point response scale ranging from 1 'never' to 5 'always') that includes a wide range of qualitative and quantitative demanding aspects of the job, such as working under time pressure, working hard, strenuous work and job complexity. The psychological job demands scale has been well-validated in Dutch samples of health care professionals (e.g. de Jonge et al., 1993; de Jonge, 1995; de Rijk et al., 1998). An example item is: "In the unit where I work, work is carried out under pressure of time". (De Jonge et al., 1999)

Physical demands are also assessed by an eight-item questionnaire with a five-point response scale ranging from 1 'never' to 5 'always'. This scale is developed by de Jonge et al. (1993) and contains items about carrying heavy loads, constrained standing, to stoop deeply and to carry shoulder high. For instance, "In my work I have to carry shoulder high for a long time". A pilot study among 249 health care workers showed an unidimensional construct with a Cronbach's alpha of 0.91 (de Jonge et al., 1993). (De Jonge et al., 1999)

Emotional demands are measured by a four-item questionnaire (five-point response scale ranging from 1 'never' to 5 'always'), which contains items about death, sickness and human suffering, aggressive and awkward patients or clients and social problems. An example is: "In my work, death, sickness or human suffering are very demanding"." (De Jonge et al., 1999)

A.5 Definitions of job resources

"Social support. Perceptions of social support in the workplace are assessed using two scales adapted from Ducharme and Martin (2000).Affective support items ask respondents to assess the following statements:" my co-workers in the NICU really care about me," "I feel close to my co-workers," "my co-workers take a personal interest in me," "I feel appreciated by my co-workers," and "my co-workers are friendly towards me" (=0.91). Five items gauge respondents' perceptions of instrumental support: "my co-

workers would fill in for me when I'm absent," "my co-workers are helpful in getting my job done," "my co-workers give useful advice on job problems," "my co-workers assist me with unusual work problems," and "my co-workers will pitch-in and help me on the job" (=0.80). Response categories for all support items range from (1) strongly disagree to (4) strongly agree, and we averaged items across each battery to create mean scales for affective and instrumental social support, respectively." (Grace and VanHeuvelen, 2019)

"Autonomy. A single measure adapted from Wrightetal.(2003) captures respondents' perceptions of work autonomy. This item asks, "How satisfied are you with the following: The chance to try your own approaches to working with patients?" Responses range from (1) very dissatisfied,to(5)very satisfied." (Grace and VanHeuvelen, 2019)

Job support: "These measure takes into account support from supervisors (5 items) and co-workers (6 items), and is designed to assess useful social interactions at work (My supervisor is concerned about the welfare of those under him; The people I work with are interested in me). Items were scored on a four-point Likert type scale, from 1 (strongly disagree) to 4 (strongly agree). High scores indicated high levels of support at work." (Blanch, 2016)

"Job autonomy is assessed by the Maastricht Autonomy Questionnaire (MAQ; de Jonge et al., 1994; de Jonge, 1995), which consists of ten Likert-items with a five-point response scale ranging from 1 'very little' to 5 'very much'. The MAQ measures the worker's opportunity or freedom, inherent in the job, to determine a variety of task elements, like the method of working, the pace of work and the work goals. The results of studies among Dutch employees confirmed the validity as well as the reliability of the MAQ (cf. de Jonge et al., 1994; de Jonge, 1995; Mulder et al., 1997)." (De Jonge et al., 1999)

"We tested the moderating effects of perceived autonomy support from the HMO and individual control over the job setting. We defined autonomy support as the extent to which physicians perceived themselves as free to maintain their individual perspectives and exercise personal choices in accomplishing clinical tasks without being inappropriately pressured (Deci and Ryan, 2011). (...) We measured perceived autonomy support using an instrument described previously (Waddimba et al., 2010, 2017). Briefly, candidate items from the Work Climate Questionnaire (Baardetal.,2004), Job Diagnostic Survey (HackmanandOldham,1980), plus new items specially created for the parent study and reworded to fit the context, underwent expert panel analysis, pilot tests, exploratory factor analysis, confirmatory factor analysis, and multi-trait analysis to construct a scale for perceived autonomy support from the HMO (Cronbach's $\alpha = 0.87$; $p < .0001$). The scale comprises six items (e.g. The HMO" interferes with how I care for patients";—"has confidence in my ability to deliver high-quality care")." (Waddimba et al., 2019)

A.6 Psychosocial work stressors

"Self-reported psychosocial work stressors were measured at every wave of the HILDA survey from 2001 to 2015. The items and scales available in all waves of the HILDA survey were based on a variety of economic and social surveys (Summerfield et al., 2016),

and three separate measures of the psychosocial work environment, previously identified through factor analysis and structural equation modelling were constructed. These were: job demands and complexity (three items, α =0.72), job control (three items, α =0.83), and perceived job security (three items, α =0.66) (Leach et al., 2010). The effort-reward imbalance model maintains that work stressors result from a perceived inequality by workers between their efforts at work and a range of rewards (Siegrist, 1996), but only a single item in HILDA assessed financial rewards relative to effort was available for assessment of any aspects of effort-reward imbalance. These psychosocial work measures have strong associations with more widely used measures including high job demands, low job control, low job security, and effort-reward unfairness (Butterworth et al., 2011; Leach et al., 2010)." (Taouk et al., 2020)